

Shri G. S. Institute of Technology and Science Indore
B.Tech. I Year Branch: Mechanical Engineering
IT10007: Fundamentals of Information Technology and Artificial Intelligence
Lab Assignment - 6

1.	Write a program to initialize a one-dimensional integer array and find the largest and smallest element in the array.
2.	Write a program to count the number of occurrences of a given element (entered by the user) in an array.
3.	Write a program to count the number of even and odd numbers in an array.
4.	Write a program to read two matrices A and B (two-dimensional array of size m x n) and print the following: (a) A + B (b) A – B (c) A x B
5.	Write a program to read two strings using fgets function and perform following operations and display the output: 1. copy one string to another using strcpy() 2. concatenate two strings using strcat() 3. compare two strings using strcmp() 4. find the length of the strings using strlen()
6.	Write a program to read two-character string arrays and perform following operations: 1. find the length of the strings without using strlen() 2. copy one string to another without using strcpy() 3. compare two strings without using strcmp()
7.	Write a program to calculate the sum of first n odd numbers using function.
8.	Write a program that defines a function countDigits(int n) which counts and returns the number of digits in a number. Example: Input: 12345 Output: 5
9.	Write a program using a function to generate Fibonacci series up to n terms. The value of n to be passed to the function.
10.	Write a program using a function to search an element in an array. Pass array and element to be searched (as entered by the user) as parameters to the function.